EGYPTIAN STOCK EXCHANGE

Data Analysis Project

PHASE 1

Research & Requirement Gathering

Based on SDLC Methodology — Data Analysis Focus

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Data collected via Yahoo Finance API & EGX Official Reports

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1. Introduction

Financial markets generate enormous volumes of data daily. Transforming this raw data into actionable intelligence is the cornerstone of modern investment strategy and academic research.

The Egyptian Stock Exchange (EGX), established in 1883, is one of the oldest and most significant stock exchanges in the Middle East and Africa. As Egypt's primary capital market, it hosts hundreds of listed companies spanning banking, real estate, telecommunications, manufacturing, and consumer goods sectors.

Data analysis applied to EGX enables investors, researchers, and policymakers to understand market dynamics, identify emerging trends, and make evidence-based decisions. This project leverages modern analytical tools and real-time API data collection to produce meaningful insights from EGX historical and live market data.

Why EGX? The exchange reflects Egypt's macroeconomic health, fiscal policy changes, and regional geopolitical developments, making it a rich and nuanced dataset for data science exploration. Additionally, EGX data is accessible through APIs such as Yahoo Finance and Alpha Vantage, enabling automated and reproducible data pipelines.

2. Problem Statement

Raw stock market data in its unprocessed form — thousands of rows of prices, volumes, and timestamps — provides little intuitive value to most stakeholders. Several critical challenges exist:

- **Data Overload:** EGX generates hundreds of thousands of data points daily across all listed securities, making manual interpretation impractical.
- **Lack of Context:** Individual price figures are meaningless without comparative benchmarks, historical context, or sector-level analysis.
- **Data Quality Issues:** Raw data frequently contains missing values, inconsistent formatting, outliers, and duplicate entries that distort analysis.
- **Accessibility Gap:** Non-technical investors lack tools to interpret complex financial time series, leaving them dependent on broker opinions.
- **Limited Forecasting:** Without systematic modelling, it is difficult to anticipate price movements, assess volatility, or predict sector performance trends.

This project addresses these challenges by building a structured analytical framework that transforms raw EGX data into clean, visualised, and interpretable insights through systematic data engineering and machine learning techniques.

3. Project Objectives

The following objectives define the analytical scope of this project across all SDLC phases:

#	Objective	Method
1	Analyze historical stock prices	Time-series analysis, pandas
2	Detect price & volume trends	Moving averages, MACD, RSI
3	Compare companies & sectors	Benchmarking, heatmaps
4	Identify top gainers & losers	Daily/weekly return ranking
5	Forecast future prices	ARIMA, Prophet, LSTM
6	Build interactive dashboards	Power BI, Matplotlib, Plotly

4. Scope of Work

This project encompasses the following analytical domains within the EGX market ecosystem:

Historical Price Data:

Daily open, high, low, close (OHLC) and adjusted close prices for EGX-listed companies spanning a minimum of 5 years.

Trading Volume Analysis:

Examination of buy/sell volume patterns, liquidity metrics, and abnormal volume detection as a market signal.

Sector Performance:

Comparative analysis across EGX sectors including Financials, Real Estate, Telecom, Energy, and Consumer Goods.

Volatility Modelling:

Standard deviation, Beta coefficient, Average True Range (ATR), and GARCH modelling for risk quantification.

Time Series Analysis:

Decomposition of trend, seasonality, and residual components; autocorrelation analysis and stationarity testing (ADF test).

API-Based Data Collection:

Automated data retrieval using Yahoo Finance API (yfinance library) and Alpha Vantage API for real-time and historical feeds.

5. Data Sources

Source	Type	Access Method	Notes
EGX Official Portal (egx.com.eg)	Historical reports, PDFs	Manual download	Quarterly & annual disclosures
Yahoo Finance API (yfinance)	OHLCV time-series	Python API (primary)	Free, real-time capable
Alpha Vantage API	Forex, stocks, indicators	REST API key	Technical indicators included
CSV / Excel Files	Structured tabular data	Local file import	Backup & offline datasets
Mubasher / Argaam	Arabic financial news & data	Web scraping / API	Regional market context

Primary data collection method: **Yahoo Finance API via the Python yfinance library**. This enables automated, reproducible, and real-time data pipelines with minimal setup, directly supporting the project's analytical and forecasting objectives.

6. Analytical Requirements

Data Cleaning & Preprocessing

- Remove duplicate rows and erroneous zero-price entries
- Standardise date formats and align trading calendars
- Normalise currency values for cross-company comparison
- Handle corporate actions: splits, dividends, rights issues

Missing Value Handling

- Forward-fill (ffill) for short gaps in time-series data
- Linear interpolation for continuous numerical fields
- Flag and document extended missing data periods
- Drop columns with greater than 30% missing values

Exploratory Data Analysis (EDA)

- Descriptive statistics: mean, median, std, skewness, kurtosis
- Distribution plots, Q-Q plots, and box plots per ticker
- Correlation matrix across selected EGX stocks
- Rolling statistics: 20-day and 50-day moving averages

Visualization

- OHLC candlestick charts with volume bars
- Sector heatmaps for weekly and monthly returns
- Interactive Plotly dashboards for dynamic exploration
- Power BI reports for executive-level summary views

Forecasting Models

- ARIMA / SARIMA: classical statistical forecasting
- Facebook Prophet: trend + seasonality decomposition
- LSTM Neural Network: deep learning for sequential data
- Model evaluation: MAE, RMSE, MAPE metrics

7. Tools & Technologies

Tool / Library	Category	Purpose
Python 3.11+	Programming Language	Core analysis & scripting
Pandas	Data Manipulation	DataFrames, time-series ops
NumPy	Numerical Computing	Array ops, statistical math
Matplotlib / Seaborn	Visualization	Static charts & plots

Plotly	Interactive Viz	Web-based dynamic charts
yfinance	API Client	Yahoo Finance data retrieval
scikit-learn	Machine Learning	Regression, clustering
statsmodels	Statistical Modelling	ARIMA, ADF test, decomp
TensorFlow / Keras	Deep Learning	LSTM forecasting models
Power BI	BI Dashboard	Executive dashboards
SQL / SQLite	Database	Data storage & querying
Jupyter Notebook	IDE	Interactive development

8. Expected Outputs

01 **Analytical Charts**
Candlestick OHLC charts, moving average overlays, volume histograms, and return distribution plots for all selected EGX securities.

02 **Interactive Dashboards**
Power BI and Plotly dashboards displaying sector performance, portfolio comparisons, and real-time price feeds via API.

03 **EDA Insights Report**
A comprehensive written report summarising statistical findings, correlations, anomalies, and market patterns detected during EDA.

04 **Forecasting Results**
Price predictions for selected stocks over 30/60/90-day horizons, accompanied by confidence intervals and model accuracy metrics.

05 **Top Performers List**
Weekly and monthly ranked lists of top gaining and losing EGX stocks by absolute return, percentage return, and risk-adjusted return.

06 **Clean Dataset**
Processed, validated, and well-documented CSV/SQLite dataset of EGX historical data, ready for future modelling or publication.

9. Risks & Challenges

Risk	Severity	Mitigation Strategy
Missing & Incomplete Data	High	Forward-fill interpolation; cross-validate against multiple sources
Inaccurate / Stale Data	Medium	Data validation pipelines; outlier detection via IQR and Z-score
Market Volatility	High	Use volatility-adjusted models; include risk metrics in outputs
API Rate Limits	Medium	Implement caching; schedule batch downloads during off-peak hours
Limited Real-Time Data	Medium	Combine historical API data with official EGX delayed feeds
Model Overfitting	Medium	Cross-validation, train/test splits, regularisation techniques

10. SDLC Roadmap

This project follows the Software Development Life Cycle (SDLC) methodology, adapted for a data analysis context. The six phases ensure a structured, repeatable, and auditable workflow:

1	**Research & Requirement Gathering ← CURRENT PHASE** Define objectives, identify data sources, plan analytical approach.
2	**Analysis** Detailed EDA, statistical profiling, correlation studies, and feature engineering to understand data characteristics.
3	**Design** Architect data pipelines, dashboard layouts, and modelling frameworks before implementation begins.
4	**Implementation** Build cleaning scripts, analytical notebooks, forecasting models, and Power BI dashboards using Python and SQL.
5	**Testing** Validate model accuracy, test data pipeline reliability, review dashboard outputs against raw data for consistency.
6	**Maintenance** Schedule automated data refresh, monitor model drift, and update dashboards as new EGX data becomes available.

11. Conclusion

This Phase 1 document establishes a comprehensive foundation for the Egyptian Stock Exchange (EGX) Data Analysis Project. Through systematic requirement gathering aligned with SDLC principles, the project has defined clear objectives, identified reliable data sources — with Yahoo Finance API as the primary collection mechanism — and outlined the analytical methods, tools, and expected deliverables.

Data-driven analysis of the EGX empowers a wide range of stakeholders: retail investors gain transparency into market performance; institutional analysts benefit from automated, reproducible pipelines; researchers access a structured dataset for academic inquiry; and policymakers receive evidence-based market health indicators.

The subsequent phases — Analysis, Design, Implementation, Testing, and Maintenance — will translate these requirements into a fully operational analytical system, producing dashboards, forecasting models, and insight reports that add genuine value to understanding one of Africa's most significant capital markets.

This document is submitted as part of the academic requirements for the Data Analysis course. All data will be collected ethically via public APIs and official EGX sources. No proprietary or confidential information is used in this project.

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